Cooperative Active Aircraft Protection Guidance Using Line-of-Sight Approach

SHASHI RANJAN KUMAR
Indian Institute of Technology Bombay, Mumbai, India

DWAIPAYAN MUKHERJEE
Indian Institute of Technology Bombay, Mumbai, India

This article proposes sliding-mode-control-based guidance strategies for a team comprised of aircraft and a defending interceptor against an attacker. The proposed guidance strategies are based on the line-of-sight guidance concept. Problems such as performance degradation under large heading angle errors, arising from linearization of engagement dynamics, are circumvented due to the nonlinear framework considered here. We show that knowledge of the bound on attacker acceleration suffices to ensure interception of an attacking interceptor. Cooperative strategies are derived for the aircraft–defender team, corresponding to cost functions based on the ℓ_p norm, using static optimization at each time instant. Simulations validate the efficacy of the proposed strategies.

Manuscript received March 1, 2019; revised December 3, 2019 and September 13, 2020; released for publication October 12, 2020. Date of publication December 21, 2020; date of current version April 10, 2021.

DOI. No. 10.1109/TAES.2020.3046328

Refereeing of this contribution was handled by A. Tsourdos.

The work of S. R. Kumar was supported in part by the Seed Grant, offered by the IRCC, IIT Bombay under Grant RD/0517-IRCCSH0-084. The work of D. Mukherjee was supported in part by the Seed Grant, offered by the IRCC, IIT Bombay under Grant RD/0518-IRCCSH0-024.

Authors' addresses: Shashi Ranjan Kumar is with the Department of Aerospace Engineering, Indian Institute of Technology Bombay, Mumbai 400076, India, E-mail: (srk@aero.iitb.ac.in); Dwaipayan Mukherjee is with the Department of Electrical Engineering, Indian Institute of Technology Bombay, Mumbai 400076, India, E-mail: (dm@ee.iitb.ac.in). *(Corresponding author: Shashi Ranjan Kumar.)*

0018-9251 © 2020 IEEE

I. INTRODUCTION

In recent years, there have been significant advancements in interceptor technology that pose serious challenges to civilian as well as military vehicles. Countermeasures, such as flares and chaff, are often used by aircraft to decoy the attackers and perform evasive maneuvers, which subsequently enhance the chances of aircraft's survival. However, these countermeasures may prove inadequate against highly sophisticated interceptors. An active aircraft protection scheme, where the aircraft launches a defender interceptor against the attacker, while cooperating with this defender, may be used to address such challenges.

Over the last decade, researchers have turned their attention significantly toward the problem of guidance design for multibody engagements. An early effort in this direction was made in [1], which presented closed-form relations while assuming constant bearing collision course. Apart from simplifying a condition in [1], Shneydor [2] also presented some relations in compact forms. A closed-form expression for the interception point in the target-centered reference frame was later presented in [3]. Furthermore, Boyell [3] also analyzed the behavior of range and bearing with engagement geometry. Yamasaki and Balakrishnan [4] observed that if the defender always remains on the attacker–target line-of-sight (LOS), then the attacker will be intercepted before reaching within a vicinity of the target (aircraft), and designed a guidance strategy based on this observation. Using optimal control theory and velocity error feedback, respectively, Yamasaki *et al.* [5] and Yamasaki and Balakrishnan [6] modified command-to-LOS (CLOS) guidance laws. However, the guidance strategies in [1]–[6] do not consider any target–defender cooperation.

The use of LOS guidance as a potential guidance strategy for the defender was investigated in [7]. To derive kinematic results, the authors considered the aircraft to be a moving launch platform and devised a cooperative strategy for the aircraft that maximizes the interceptor-to-defender lateral acceleration ratio. Ratnoo and Shima [8] analyzed a given set of attacking interceptor and defender guidance strategies and analytically obtained the launch envelopes, acceleration ratios of the adversaries, and closed-form expressions for interceptor launch conditions that enable the interceptor to evade the defender.

Linear–quadratic differential game theory was used in [9]–[13] to devise guidance strategies for three-body engagements. The problem was formulated as a dynamic game between two adversarial teams in [9] and [10]. The solutions of these games were then obtained as a numerical solution of the Riccati equation that uses impulse functions in the indices to reflect disappearance of the defenders after their capture by the attacking interceptor. In [11], a linearized engagement dynamics was used to derive algebraic conditions for the pursuer to capture the evader, while evading the defender with bounded control. In [12], Rusnak *et al.* show that when the weight on the defender's control effort tended to zero, the guidance commands of the interceptor and the target were identical to those obtained in

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

a game in which the defender was absent. Cooperative evasion and pursuit strategies, for the target and the defender, respectively, were derived in [13] along with an optimal strategy for the attacking interceptor to intercept the target while successfully evading the defender. The adversaries were assumed to have linear dynamics of arbitrary order.

Guidance strategies for three-body engagements were designed in [14]–[19] using optimal control theory. A cooperative guidance strategy, which seeks to maximize the separation between an attacking interceptor and the target, was proposed in [14] and [15]. However, the method in [14] and [15] may be computationally intensive, since the guidance command in [14] was obtained as the solution of a two-point boundary value problem, and that in [15] necessitated obtaining roots of an eighth-order polynomial. Cooperative guidance strategies for both the defender and the target, with arbitrary order linear dynamics, were presented in [16], when the aircraft–defender team had information about the class of linear guidance strategies adopted by the attacker. Shaferman and Shima [17] proposed a cooperative guidance and estimation scheme, where the cooperative estimator is a nonlinear adaptation of multiple-model adaptive estimator. Every model represents a possible guidance strategy and guidance parameters of the attacking interceptor. The cooperation arises from fact that the defender is able to anticipate the induced maneuver on the interceptor through its knowledge of the target's maneuver. Cooperative guidance strategies were presented in [18] corresponding to one-way and two-way cooperations between the target and the defender. In [19], Weiss *et al.* presented two algorithms that were derived based on the assumption that adversaries' performance can be prescribed by the required maximum miss distance for pursuit and the required minimum miss distance for evasion, while minimizing the control effort.

A majority of the guidance strategies discussed so far were presented within a linearized framework. Their performance may, therefore, degrade significantly in the presence of large heading errors, since the linearization is no longer valid. Such performance deterioration may be unacceptable in practice. Moreover, the minimization of the total control effort during the engagement, using optimal control techniques within a linearized framework, may be intractable in a nonlinear setup. A nonlinear guidance strategy for both the target and the defender was proposed in [20] using sliding-mode control (SMC) [21]. Additionally, objectives such as reduction of the sensitivity of miss distance with respect to time-to-go were also considered therein. However, linearized engagement dynamics was used to compute the zero-effort-miss in [20], and the strategy does not aim to optimize any cost function.

The main contributions of this article are enumerated as follows.

1) Guidance strategies for both the defender and aircraft are derived in a nonlinear framework using SMC, which ensures interception of the attacking interceptor. By considering the nonlinearity, problems due to linearization are eliminated. Previously, only Kumar and Shima [20] had used SMC for such three-body engagement scenarios, to the best of authors' knowledge.
2) The proposed strategies for the aircraft–defender team do not require information about attacking interceptor's guidance strategy or its actual maneuver, knowledge of the bound on attacking interceptor acceleration suffices.
3) During implementation, the time-to-go estimates are not required explicitly, unlike most of the optimal-control-based guidance strategies.
4) Different levels of cooperation between aircraft and defender are proposed based on various cost functions, and their performances are compared through simulations.
5) Guidance commands for two-way cooperation are derived using instantaneous static optimization.
6) A finite-time version of the guidance strategies is presented to guarantee the convergence of the defending interceptor on desired course, within a finite time.
7) Finally, we show that the cooperative strategies based on the optimization of aforementioned cost functions remain effective even for the finite-time version.

Some preliminary results were presented in [24].

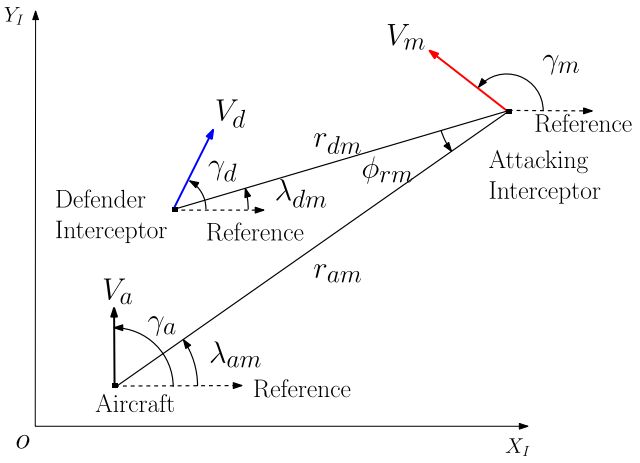


Fig. 1. Engagement geometry for three-point guidance design.

II. PROBLEM FORMULATION

Consider a planar engagement scenario, as in Fig. 1, where an attacking interceptor (denoted as "attacker" hereafter) is launched to intercept an aircraft. Since the attacker views the aircraft as its target, henceforth we shall refer to the aircraft as the "target." The aircraft (target) deploys an interceptor as a defensive measure to intercept the attacker, and this interceptor is hence called the "defender." The three players, i.e., attacker, defender, and target, are assumed to be point-mass vehicles. The coordinate axes $X_I - O - Y_I$ in Fig. 1 represent the inertial coordinate system. The velocities and flight path angles of the target, the defender, and the attacker are denoted by (V_a, V_d, V_m) and $(\gamma_a, \gamma_d, \gamma_m)$, respectively. The subscripts a, d, and m stand for aircraft,

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

defender, and attacker, respectively. The LOS angle and the relative distance of engagement geometries are denoted by λ_i and r_i, respectively, where $i = am, dm$, as shown in Fig. 1, with the subscript i indicating the pair of vehicles under consideration. We assume the constant speeds of the attacker and the defender to be comparable and higher than that of the target. In other words, $V_d \approx V_m > V_a$.

For the defender–attacker and aircraft–attacker engagements, the range rates are given by

$$\dot{r}_{dm} = V_m \cos\left(\gamma_m - \lambda_{dm}\right) - V_d \cos\left(\gamma_d - \lambda_{dm}\right) \quad (1a)$$

$$\dot{r}_{am} = V_m \cos\left(\gamma_m - \lambda_{am}\right) - V_a \cos\left(\gamma_a - \lambda_{am}\right) \quad (1b)$$

while their corresponding LOS rates are given by

$$r_{dm}\dot{\lambda}_{dm} = V_m \sin\left(\gamma_m - \lambda_{dm}\right) - V_d \sin\left(\gamma_d - \lambda_{dm}\right) \quad (2a)$$

$$r_{am}\dot{\lambda}_{am} = V_m \sin\left(\gamma_m - \lambda_{am}\right) - V_a \sin\left(\gamma_a - \lambda_{am}\right). \quad (2b)$$

Equations (1) and (2) represent the kinematics resolved along and perpendicular to the LOS, with respect to the engagement shown in Fig. 1. The heading angles of the adversaries are given by

$$\dot{\gamma}_i = \frac{a_i}{V_o}, |a_i| \leq a_i^{\max} \quad \forall i \in \{a, d, m\} \quad (3)$$

where a_i denotes the lateral accelerations of the ith adversary, which is assumed to be bounded by $a_i^{\max}$. The attacker–target–defender engagement is clearly a three-body interception problem, which we formally state as follows.

Problem: Design guidance strategies for the team comprising of the target and the defender so as to ensure interception of the attacker and protection of the target, for varying levels of target–defender cooperation.

The goal of cooperation is to optimize some measures of the overall control effort for the target–defender team. This will be discussed in some detail at a later stage. But unlike optimal-control-based strategies, that require explicit time-to-go information, in this article, we shall not endeavor to minimize the overall control energy throughout the entire duration of the engagement.

III. GUIDANCE DESIGN

In this section, we first present the dynamics of the three-body planar engagement scenario. This is followed by design of suitable guidance strategies.

LEMMA 1 The dynamics of the LOS angles are of relative degree 2 with respect to the lateral accelerations of the corresponding vehicles and are given by

$$\begin{aligned}\ddot{\lambda}_{dm} = & -\frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{\cos\left(\gamma_d - \lambda_{dm}\right)}{r_{dm}} a_d \\ & + \frac{\cos\left(\gamma_m - \lambda_{dm}\right)}{r_{dm}} a_m \\ \ddot{\lambda}_{am} = & -\frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} - \frac{\cos\left(\gamma_a - \lambda_{am}\right)}{r_{am}} a_a \\ & + \frac{\cos\left(\gamma_m - \lambda_{am}\right)}{r_{am}} a_m. \end{aligned} \quad (4)$$

PROOF Upon differentiating (2b) with respect to time, and substituting $\dot{\gamma}_i$ from (3), it follows that

$$\begin{aligned}\dot{r}_{dm}\dot{\lambda}_{dm} + r_{dm}\ddot{\lambda}_{dm} = & a_m \cos\left(\gamma_m - \lambda_{dm}\right) - \dot{r}_{dm}\dot{\lambda}_{dm} \\ & - a_d \cos\left(\gamma_d - \lambda_{dm}\right). \end{aligned} \quad (5)$$

The dynamics of λ_{dm} is, thus, of degree two with respect to both a_d and a_m. Similarly, (4) holds for the dynamics of λ_{am} too. This completes the proof. ■

Our guidance philosophy is based on the premise that if the defender can reach the LOS between the attacker and the target (aircraft), before the attacker manages to capture the target and stays on this LOS, then interception of the attacker by the defender is guaranteed before the attacker intercepts the aircraft [4], [5]. A concept similar to the one in CLOS guidance [7] is used here. This is also referred to as the triangle guidance concept in the literature [4].

To impose the aforementioned geometric constraint on the defender position, we shall use the concept of SMC. Subject to certain conditions, sliding-mode controllers maintain sliding mode on switching surface, which is usually chosen to meet certain objectives [21]. Thus, once the sliding mode is enforced on the chosen surface, the system's dynamics becomes insensitive to uncertainties and external disturbances. Note that SMC and related concepts were also shown to be useful for other aerospace applications, such as cooperative salvo [22], [23]. We define the switching surface using the angle between two LOSs, denoted by ϕ_{rm} in Fig. 1. If this angle, ϕ_{rm}, can be made zero, then the defender will lie on the LOS of aircraft and attacker and between them, as desired. As a result, before the attacker can capture the target, the defender is sure to intercept the attacker. From Fig. 1, it follows that ϕ_{rm} can be written as

$$\phi_{rm} = \lambda_{am} - \lambda_{dm}. \quad (6)$$

The following lemma describes the relation between the angle ϕ_{rm} and the defender's lateral acceleration, a_d.

LEMMA 2 The dynamics of ϕ_{rm} has a relative degree two with respect to the defender's lateral acceleration, a_d, and has a disturbance term, d, given by

$$d = \left[\frac{\cos\left(\gamma_m - \lambda_{ad}\right)}{r_{am}} - \frac{\cos\left(\gamma_m - \lambda_{dm}\right)}{r_{dm}}\right] a_m \quad (7)$$

which can be further bounded by $d_{\max}$, as follows:

$$|d| \leq d_{\max}, d_{\max} = \left(\frac{1}{r_{dm}} + \frac{1}{r_{am}}\right) a_m^{\max}. \quad (8)$$

PROOF On differentiating (6), one may obtain

$$\dot{\phi}_{rm} = \dot{\lambda}_{am} - \dot{\lambda}_{dm}. \quad (9)$$

Upon further differentiating (9), and using (4), we obtain

$$\begin{aligned}\ddot{\phi}_{rm} = & \ddot{\lambda}_{am} - \ddot{\lambda}_{dm} \\ = & -\frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} + \frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{\cos\left(\gamma_a - \lambda_{am}\right)}{r_{am}} a_a \\ & + \frac{\cos\left(\gamma_d - \lambda_{dm}\right)}{r_{dm}} a_d + d \end{aligned} \quad (10)$$

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

where the term d may be treated as an unknown disturbance, given by (7). Since a_m is assumed to be unknown to the defender, the terms associated with it can be clubbed together as an unknown disturbance, d. Thus, the dynamics of ϕ_{rm}, as given by (10), has a relative degree of 2 with respect to a_d.

Consider $d_{\max}$ to be the maximum value of disturbance, so that $|d| \leq d_{\max}$. Noting that d can be expressed as

$$\begin{aligned} |d| &= \left| \left(\frac{\cos\left(\gamma_m - \lambda_{ad}\right)}{r_{am}} - \frac{\cos\left(\gamma_m - \lambda_{dm}\right)}{r_{dm}} \right) a_m \right| \\ &\leq \left[\frac{|\cos\left(\gamma_m - \lambda_{ad}\right)|}{r_{am}} + \frac{|\cos\left(\gamma_m - \lambda_{dm}\right)|}{r_{dm}} \right] a_m^{\max} \end{aligned} \tag{11}$$

we use the fact that $|\cos(x)| \leq 1 \forall x$, to deduce $|d| \leq d_{\max}$. ■

Note that the above bound in (8) is a conservative one, and the actual upper bound could be much lesser in practice, especially when we consider a nonzero lethal radius for the warhead. To derive the guidance command using SMC theory, we define the switching surface as

$$S = \dot{\phi}_{rm} + \eta \phi_{rm}, \eta > 0. \tag{12}$$

It is evident that the switching surface in (12) ensures asymptotic convergence after occurrence of sliding mode, and a higher value of the constant term, η, results in a higher rate of convergence [21]. On differentiating (12) and substituting for $\ddot{\phi}_{rm}$ from (10), it follows that

$$\begin{aligned} \dot{S} = &\frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} - \frac{\cos\left(\gamma_a - \lambda_{am}\right)}{r_{am}} a_a \\ &+ \frac{\cos\left(\gamma_d - \lambda_{dm}\right)}{r_{dm}} a_d + d + \eta \dot{\phi}_{rm}. \end{aligned} \tag{13}$$

The levels of cooperation between the target and the defender may vary considerably. In case of one-way cooperation, the target performs its own maneuvers to evade the attacker, irrespective of any maneuvers that the defender may possibly perform. However, the target informs the defender about its own maneuver. On the other hand, in case of two-way cooperation, both the target and the defender aspire to jointly minimize some measure of their individual control components. This is aimed at limiting the individual lateral accelerations of the cooperating vehicles at every instant. We shall discuss these aspects, in some detail, in the following subsections.

A. One-Way Cooperation Between the Target and the Defender

The sliding-mode controller comprises two parts

$$a_d = a_d^{\text{eq}} + a_d^{\text{disc}} \tag{14}$$

where a_d^{eq} and a_d^{disc} represent the equivalent and discontinuous control components, respectively. We will use the equivalent control approach [21], in which the component a_d^{eq} is obtained by evaluating $\dot{S} = 0$. Note that the component, a_d^{eq}, acts even when the sliding mode is enforced on chosen switching surface. The equivalent control component, upon neglecting the disturbance term, d, may be written as

$$\begin{aligned} a_d^{\text{eq}} &= \frac{-r_{dm}}{\cos\left(\gamma_d - \lambda_{dm}\right)} \\ &\times \left[\frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} - \frac{\cos\left(\gamma_a - \lambda_{am}\right)}{r_{am}} a_a + \eta \dot{\phi}_{rm} \right] \\ &= \frac{-r_{dm}}{\cos\left(\gamma_d - \lambda_{dm}\right)} \\ &\times \left[\frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} - \frac{\cos\left(\gamma_a - \lambda_{am}\right)}{r_{am}} a_a + \eta\left(\dot{\lambda}_{am} - \dot{\lambda}_{dm}\right) \right]. \end{aligned} \tag{15}$$

Now, we may choose the discontinuous component of the sliding-mode controller as

$$a_d^{\text{disc}} = -\frac{\mathscr{M}_1}{\cos\left(\gamma_d - \lambda_{dm}\right)} \text{sign}(S), \mathscr{M}_1 > 0 \tag{16}$$

where $\mathscr{M}_1$ denotes the gain of the discontinuous part. The following result provides a guideline for choosing $\mathscr{M}_1$.

THEOREM 1 For the three-body planar engagement, whose relative dynamics is described by (4), and the lateral acceleration of defender is chosen as (14)–(16), the angular deviation ϕ_{rm} from attacker–target LOS and its derivative $\dot{\phi}_{rm}$ converge to zero for a choice of gains given by

$$\mathscr{M}_1 > \left(1 + \frac{r_{dm}}{r_{am}}\right) a_m^{\max}. \tag{17}$$

PROOF Choosing a Lyapunov function candidate, $\mathscr{V}$, as

$$\mathscr{V} = \frac{1}{2} S^2 \tag{18}$$

and subsequently differentiating (18) with respect to time, after substituting for $\dot{S}$ using (13), we obtain

$$\begin{aligned} \dot{\mathscr{V}} = S \Bigg[&\frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} - \frac{\cos\left(\gamma_a - \lambda_{am}\right)}{r_{am}} a_a \\ &+ \frac{\cos\left(\gamma_d - \lambda_{dm}\right)}{r_{dm}} a_d + d + \eta \dot{\phi}_{rm} \Bigg]. \end{aligned} \tag{19}$$

On substituting a_d using (14)–(16) into (19), and performing some algebraic manipulations, we may arrive at the following expression:

$$\begin{aligned} \dot{\mathscr{V}} &= S \left(\frac{-\mathscr{M}_1}{r_{dm}} \text{sign}(S) + d \right) \leq -|S| \left(\frac{\mathscr{M}_1}{r_{dm}} - |d| \right) \\ &\leq -|S| \left(\frac{\mathscr{M}_1}{r_{dm}} - d_{\max} \right). \end{aligned} \tag{20}$$

If we choose the gain, $\mathscr{M}_1$, to satisfy

$$\mathscr{M}_1 > r_{dm} d_{\max} \tag{21}$$

then the derivative of the Lyapunov function, $\dot{\mathscr{V}}$, is always negative definite, leading to the occurrence of sliding mode within a finite time. Consequently, the terms ϕ_{rm} and its rate

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

$\dot{\phi}_{rm}$ converge to zero. The objective of the guidance law is, thus, satisfied from the defender's point of view. From (21), and (8), the bound on gain $\mathscr{M}_1$ can be derived as in (17). This completes the proof. ■

We may rewrite the equivalent component of the guidance command, given in (15), after recollecting similar terms as

$$a_d^{\text{eq}} = -\left(\frac{2\dot{r}_{dm} - \eta r_{dm}}{\cos\left(\gamma_d - \lambda_{dm}\right)}\right)\dot{\lambda}_{dm} + \frac{r_{dm}}{r_{am}}\left(\frac{2\dot{r}_{am} - \eta r_{am}}{\cos\left(\gamma_d - \lambda_{dm}\right)}\right)\dot{\lambda}_{am} + \frac{r_{dm}}{r_{am}}\frac{\cos\left(\gamma_a - \lambda_{am}\right)}{\cos\left(\gamma_d - \lambda_{dm}\right)}a_a. \tag{22}$$

The last two terms of (22) depend on the information provided by the aircraft to the defender. Additionally, they are directly proportional to the ratio of attacker–defender distance to the attacker–aircraft distance, which is (r_{dm}/r_{am}). The defender assigns full weightage to the aircraft information at the time of launch when $(r_{dm}/r_{am}) = 1$. This is evident from (22). As the defender's distance from the aircraft increases, and the defender closes in on the attacker, (r_{dm}/r_{am}) decreases and is much smaller than unity. At this juncture, the defender mainly relies on its own information and assigns less weightage to the information provided by the aircraft.

Under the assumption that the time-to-go is approximated by its conventional estimation given by $t_{\text{go}}^{dm} = -r_{dm}/\dot{r}_{dm}$, $t_{\text{go}}^{am} = -r_{am}/\dot{r}_{am}$, the equivalent control (22) further reduces to

$$a_d^{\text{eq}} = \frac{(2 + \eta t_{\text{go}}^{dm})V_{c_{dm}}\dot{\lambda}_{dm}}{\cos\left(\gamma_d - \lambda_{dm}\right)} - \frac{r_{dm}}{r_{am}}\frac{(2 + \eta t_{\text{go}}^{am})V_{c_{am}}\dot{\lambda}_{am}}{\cos\left(\gamma_d - \lambda_{dm}\right)} + \frac{r_{dm}}{r_{am}}\frac{\cos\left(\gamma_a - \lambda_{am}\right)}{\cos\left(\gamma_d - \lambda_{dm}\right)}a_a \tag{23}$$

where $V_{c_{dm}}$ and $V_{c_{am}}$ are the closing velocities of the defender–attacker and aircraft–attacker engagements, respectively. This further elucidates the dependence of the weights assigned by the defender to the aircraft's information, as the distance of the defender from the attacker varies. As $r_{dm} \to 0$, so does t_{go}^{dm}. Consequently, $a_d^{\text{eq}} \to (2V_{c_{dm}}\dot{\lambda}_{dm})/\cos(\gamma_d - \lambda_{dm})$, which is similar to proportional navigation (PN) class of guidance strategies.

B. Two-Way Cooperation Between the Target and the Defender

When there is a two-way cooperation between the target and the defender, it opens up the possibility of reducing the individual control efforts. This, in turn, may prevent the accelerations from reaching their saturation limits. Even if the lateral accelerations saturate at some instant of time, cooperation may facilitate the target–defender team to pull out of their maximum accelerations sooner. From (13), we may conclude that the target and defender lateral accelerations must satisfy the following condition, in order to enforce sliding mode on (12):

$$a_1 a_a - a_2 a_d = \mathscr{F} \tag{24}$$

where the coefficients a_1, a_2, and the term $\mathscr{F}$ are defined as

$$a_1 = \frac{\cos\left(\gamma_a - \lambda_{am}\right)}{r_{am}},\ a_2 = \frac{\cos\left(\gamma_d - \lambda_{dm}\right)}{r_{dm}}$$
$$\mathscr{F} = \frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} + \eta\dot{\phi}_{rm} + \mathscr{M}_2\text{sign}(S). \tag{25}$$

The guidance design involves choice of gain $\mathscr{M}_2$, and the following result serves to aid us along that direction.

THEOREM 2 For the three-body planar engagement, whose relative dynamics is described by (4), and the lateral accelerations of both the target and the defender are chosen such that they satisfy the constraint equation (24), the angular deviation ϕ_{rm} and its derivative $\dot{\phi}_{rm}$ converge to zero for a choice of gain given by

$$\mathscr{M}_2 > \left(\frac{1}{r_{dm}} + \frac{1}{r_{am}}\right)a_m^{\max}. \tag{26}$$

PROOF Consider the same Lyapunov function candidate, $\mathscr{V}$, as defined in (18). As shown earlier, its derivative is given by (19). Upon imposing the constraint on a_d and a_a from (24) in (19), and performing some algebraic manipulations, we obtain

$$\dot{\mathscr{V}} = S\left[-\mathscr{M}_2\text{sign}(S) + d\right] \le -|S|\left(\mathscr{M}_2 - d_{\max}\right). \tag{27}$$

If we choose the gain, $\mathscr{M}_2$, to satisfy

$$\mathscr{M}_2 > d_{\max} \tag{28}$$

where $d_{\max}$ is the same as in (8), the occurrence of the sliding mode in a finite time can be ensured owing to the negative definiteness of $\dot{\mathscr{V}}$. Thus, ϕ_{rm} and its derivative converge to zero, and the target–defender team achieves successful interception in a cooperative fashion. Using (28), and (8), we obtain the bound on gain $\mathscr{M}_2$, as given by (26). ■

From (21) and (28), we may deduce that the lower bound on $\mathscr{M}_1$ is higher than that of $\mathscr{M}_2$, since $r_{dm} > 1$ for most part of the engagement and the lethal radius of the warhead, r_{lethal}, is normally greater than or equal to 1 m. This reduction in gain requirement is an advantage of employing cooperation between the aircraft and the defender.

At each time instant, the straight line in the a_d–a_a space, described by (24), constrains the feasible lateral acceleration values of the target and the defender, for a successful interception of the attacker. Evidently, there are infinitely many choices for a_a and a_d that satisfy (24). Since there are two degrees of freedom, apart from the successful interception of the attacker by the target–defender team, we may also accomplish additional objectives.

To minimize the control efforts, we may reformulate the main problem in Section II as a minimization of the following cost function:

$$\boldsymbol{X}^{\star} = \arg\min J,\ J = \|\boldsymbol{X}\|,\ \boldsymbol{X} = \begin{bmatrix} \dfrac{a_d}{w_d} & \dfrac{a_a}{w_a} \end{bmatrix}^T \tag{29}$$

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

subject to the constraint given by (24), where $\|.\|$ is some norm in $\mathbb{R}^2$ at an instant of time. The terms w_a and w_d denote the positive weights assigned to the lateral accelerations of the aircraft and the defender, respectively, during optimization. In this article, we particularly consider the minimizations of the following objective functions:

$$J_p = \|\boldsymbol{X}\|_p = \left[\left|\frac{\mathscr{F}+a_2a_d}{a_1w_a}\right|^p + \left|\frac{a_d}{w_d}\right|^p\right]^{1/p} \tag{30a}$$

$$J_2 = \|\boldsymbol{X}\|_2 = \left[\left(\frac{\mathscr{F}+a_2a_d}{a_1w_a}\right)^2 + \left(\frac{a_d}{w_d}\right)^2\right]^{1/2} \tag{30b}$$

$$J_1 = \|\boldsymbol{X}\|_1 = \left|\frac{\mathscr{F}+a_2a_d}{a_1w_a}\right| + \left|\frac{a_d}{w_d}\right| \tag{30c}$$

$$J_\infty = \|\boldsymbol{X}\|_\infty = \max\left[\left|\frac{a_a}{w_a}\right|, \left|\frac{a_d}{w_d}\right|\right] \tag{30d}$$

subject to the constraint (24).

The cost functions in (30) aim to minimize the instantaneous accelerations rather than minimizing the total control effort during entire duration of engagement. Thus, the problem is one of static optimization rather than a dynamic one.

In case we choose $w_d = a_d^{\max}$ and $w_a = a_a^{\max}$, (30d) will have a feasible solution only when the accelerations of the defender and the aircraft are within their respective bounds. Now, we obtain optimal values of the cost functions in (30).

THEOREM 3 The cost function in (30a), subject to constraint (24), is minimized when the lateral accelerations of the defender and the aircraft are chosen as

$$a_d = -\frac{\mathscr{F}\text{sign}(a_2)|a_2|^{1/(p-1)}}{|a_2|^{p/(p-1)} + |a_1\Gamma|^{p/(p-1)}};\; a_a = \frac{\mathscr{F}+a_2a_d}{a_1} \tag{31}$$

with $\Gamma = w_a/w_d$ and integral values of $p > 1$.

PROOF On differentiating J_p with respect to a_d, one obtains

$$\frac{\partial J_p}{\partial a_d} = \frac{1}{pJ_p^{p-1}}\left[\frac{pa_2|\mathscr{F}+a_2a_d|^{p-1}\text{sign}(\mathscr{F}+a_2a_d)}{|a_1w_a|^p} + \frac{p|a_d|^{p-1}\text{sign}(a_d)}{|w_d|^p}\right]. \tag{32}$$

Furthermore, the solution of $\dfrac{\partial J_p}{\partial a_d} = 0$ results in

$$\frac{a_2|\mathscr{F}+a_2a_d|^{p-1}\text{sign}(\mathscr{F}+a_2a_d)}{|a_1w_a|^p} = -\frac{|a_d|^{p-1}\text{sign}(a_d)}{|w_d|^p}. \tag{33}$$

This can be transformed to the following equation:

$$\left|\frac{\mathscr{F}}{a_d}+a_2\right|^{p-1}\text{sign}\left(\frac{\mathscr{F}}{a_d}+a_2\right) = -\frac{1}{a_2}\left|\frac{a_1w_a}{w_d}\right|^p = -\frac{|a_1\Gamma|^p}{a_2}. \tag{34}$$

With $x := a_2 + \mathscr{F}/a_d$ and $K := -|a_1\Gamma|^p/a_2$, (34) reduces to

$$|x|^{p-1}\,\text{sign}(x) = K. \tag{35}$$

Let $p > 1$ be an even integer. As $p-1$ is an odd integer, (35) reduces to $x^{p-1} = K$. The solution to this equation is given by

$$x = K^{1/(p-1)} = |K|^{1/(p-1)}\text{sign}(K). \tag{36}$$

For the case when $p > 1$ is an odd integer, we have

$$|x|^{p-1}\,\text{sign}(x) = |K|\text{sign}(K). \tag{37}$$

Now, since the $(p-1)$th root exists only if $\text{sign}(K) = \text{sign}(x)$, the solution in this case is given by

$$x = |K|^{1/(p-1)}\text{sign}(K). \tag{38}$$

The solutions of x, for both odd and even integer values of p, are identical, which on substituting for x and K results in

$$\frac{\mathscr{F}}{a_d}+a_2 = \left|-\frac{|a_1\Gamma|^p}{a_2}\right|^{1/(p-1)}\text{sign}\left(-\frac{|a_1\Gamma|^p}{a_2}\right) = -\left|\frac{|a_1\Gamma|^p}{a_2}\right|^{1/(p-1)}\text{sign}\,(a_2)\,. \tag{39}$$

We obtain defender lateral acceleration as

$$a_d = -\frac{\mathscr{F}}{a_2 + \left|\frac{|a_1\Gamma|^p}{a_2}\right|^{1/(p-1)}\text{sign}\,(a_2)} \tag{40}$$

which on simplifying and using (24) leads to guidance commands, as given in (31). This completes the proof. ■

In order to glean further geometric interpretations and insights about the optimization procedure, we now focus on minimization of J_2.

COROLLARY 1 The cost function in (30b), subject to constraint (24), is minimized when the defender and aircraft lateral accelerations are, respectively, chosen as

$$a_d = -\frac{a_2\mathscr{F}}{a_2^2+a_1^2\Gamma^2}, a_a = \frac{a_1\Gamma^2\mathscr{F}}{a_2^2+a_1^2\Gamma^2}. \tag{41}$$

PROOF Using Theorem 3, the lateral accelerations for both the defender and the aircraft corresponding to the optimization of J_2 can be obtained by substituting $p = 2$ into (31), which leads to (41), and that concludes the proof. ■

A geometric interpretation for the optimization of J_2 can be obtained by considering the level set for $J_2 = c > 0$, which is an ellipse in the a_d–a_a plane. As described by (24), the constraint equation represents a straight line in the control space. Hence, the optimum solution corresponding to J_2 is the point where the constraint line is tangent to the elliptical contour of $J_2 = c^\star$, with $c^\star$ being the minima of J_2. This is illustrated in Fig. 2(a).

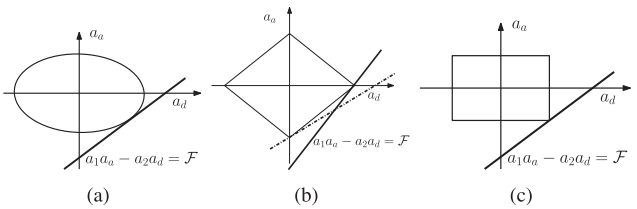


Fig. 2. Minimization of cost function in control space [(a_d, a_a) space]. (a) ℓ_2 norm. (b) ℓ_1 norm. (c) ℓ_∞ norm.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

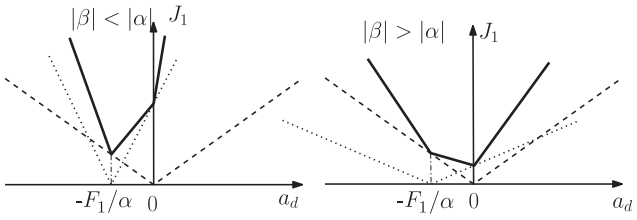


Fig. 3. Minimizations of the cost function J_1.

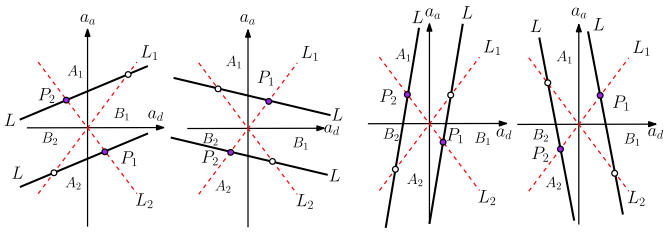


Fig. 4. Minimizations of the cost function J_∞. (a) Illustration for zone A. (b) Illustration for zone B.

The cost function J_1 can be rewritten, using (24), as $J_1 = |\mathscr{F}_1 + \alpha a_d| + |\beta a_d|$, where $\alpha = a_2/(a_1 w_a)$, $\beta = 1/w_d$, $\mathscr{F}_1 = \mathscr{F}/(a_1 w_a)$.

THEOREM 4 The cost function in (30c), subject to constraint (24), is minimized when the defender and aircraft lateral acceleration pair is chosen as

$$(a_d, a_a) = \begin{cases} \left(-\dfrac{\mathscr{F}}{a_2}, 0\right) & \left|\dfrac{a_2}{a_1}\right| > \Gamma \\ \left(0, \dfrac{\mathscr{F}}{a_1}\right) & \left|\dfrac{a_2}{a_1}\right| < \Gamma. \end{cases} \tag{42}$$

PROOF Variation of the cost function for the cases, where $|\alpha| > |\beta|$ and $|\alpha| < |\beta|$, is shown in Fig. 3. In Fig. 3, observe that for $|\alpha| > |\beta|$, the minima of J_2 is obtained for $a_d = -\mathscr{F}_1/\alpha = -\mathscr{F}/a_2$, while for $|\alpha| < |\beta|$, it occurs at $a_d = 0$. Furthermore, observe that $|\alpha| > |\beta|$ implies $|a_2/a_1| > \Gamma$, and $|a_2/a_1| < \Gamma$ is equivalent to $|\alpha| < |\beta|$. ■

The level sets of J_1 are rhombuses, as shown in Fig. 2(b). From Fig. 2(b), it follows that if the absolute value of the slope of constraint line, $|a_2/a_1|$, is greater than Γ, then the solution will be the intersection of rhombus-shaped contour with the constraint line in (24) at the vertex on the a_d-axis. However, if $|a_2/a_1| < \Gamma$, the solution will be given by the vertex of the rhombus on the a_a-axis.

If $|a_2/a_1| = \Gamma$, then there are infinitely many solutions to the optimization problem in (30c). In fact, any point on the constraint line in the control space, between the x and y intercepts of (24), qualifies as a solution to this problem.

Note that to minimize the cost function J_∞, the solution is such that only one control, that is, the lateral acceleration of either aircraft or defender, is effective at any time instant.

THEOREM 5 The cost function in (30d), subject to constraint (24), is minimized when the defender and aircraft lateral acceleration pair is chosen as

$$(a_d, a_a) = \begin{cases} \left(\dfrac{-\mathscr{F}}{\Gamma a_1 + a_2}, \dfrac{\Gamma \mathscr{F}}{\Gamma a_1 + a_2}\right) & \dfrac{a_2}{a_1} > 0 \\ \left(\dfrac{\mathscr{F}}{\Gamma a_1 - a_2}, \dfrac{\Gamma \mathscr{F}}{\Gamma a_1 - a_2}\right) & \dfrac{a_2}{a_1} < 0. \end{cases} \tag{43}$$

PROOF Consider the four zones A_1, A_2, B_1, and B_2, into which the control space is partitioned by the straight lines L_1 and L_2, given by $a_a = \pm \Gamma a_d$, as shown in Fig. 4. For zones A_1 and A_2, $|a_a/w_a| > |a_d/w_d|$ and *vice versa* for zones B_1 and B_2. Consequently, the ∞-norms, for the pairs of zones (A_1, A_2) and (B_1, B_2), correspond to $|a_a/w_a|$ and $|a_d/w_d|$, respectively, which need to be minimized in order to compute the minima of the cost function in (30d). The constraint line of (24), denoted by L, intersects these two lines, L_1 and L_2, at two different points, unless L is parallel to either of them. Depending on the nature of intersection, the solution will lie in the appropriate zone, as shown in Fig. 4. Furthermore, the constraint line will dictate if the solution is obtained at point P_1 or P_2. If $(a_2/a_1) > 0$, then P_1 and P_2 are obtained as points of intersection of L and L_2 (which has a negative slope); otherwise, the intersection of L and L_1 (the line with positive slope) yields the solution. For the positive slope of constraint line, that is, $(a_2/a_1) > 0$, the guidance commands take the form of

$$a_d = \frac{-\mathscr{F}}{\Gamma a_1 + a_2}, a_a = \frac{\Gamma \mathscr{F}}{\Gamma a_1 + a_2}. \tag{44}$$

Similarly, for negative slope of constraint line, that is, $(a_2/a_1) < 0$, the guidance commands are given by

$$a_d = \frac{\mathscr{F}}{\Gamma a_1 - a_2}, a_a = \frac{\Gamma \mathscr{F}}{\Gamma a_1 - a_2}. \tag{45}$$

This completes the proof. ■

The level sets of J_∞ are rectangles, as shown in Fig. 2(c). From Fig. 2(c), it follows that if the slope of the constraint line, (a_2/a_1), is positive, then the solution will correspond to the intersection of the vertices of the rectangle on the second or fourth quadrant with the constraint line (24). Similarly, for $a_2/a_1 < 0$, the solution will correspond to the intersection of the vertices of the rectangle in the first or the third quadrant with the constraint line.

If a_1 or a_2 is zero, the slope of the constraint line is ∞ or 0, respectively. In such cases, the optimization problem will have an infinite number of solutions that lie on the segment of the constraint line in zone A_i (when $a_2 = 0$) or B_i (when $a_1 = 0$), where $i \in \{1, 2\}$. The analysis in Section III-B holds even for such cases where $\mathscr{F}$ may differ from the one in (25) for other techniques that ensure successful interception.

The time taken by the defender to reach the LOS must be less than that required by the attacker to intercept the aircraft. This can be achieved by ensuring finite-time convergence of the controlled variable, ϕ_{rm}, to zero.

If the finite-time convergent switching surface is chosen for convergence of defender on LOS between the aircraft

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

and the attacker, then the sliding surface would be of the form

$$\sigma = \phi_{rm} + \frac{1}{\zeta}\dot{\phi}_{rm}^{\xi}, \zeta > 0, 1 < \xi = \frac{m}{n} < 2 \qquad (46)$$

where both m and n are odd integers. The corresponding expression for $\mathscr{F}$ in (24), for this case, is given by

$$\mathscr{F} = \frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}} + \frac{\dot{\phi}_{rm}^{2-\xi}\zeta}{\xi} + \frac{\mathscr{M}_3\zeta}{\xi}\text{sign}(\sigma). \qquad (47)$$

With $\mathscr{F}$ chosen as in (47), $\dot{\sigma} = d + (\mathscr{M}_3\xi/\zeta)\dot{\phi}_{rm}^{\xi-1}$ sign(σ). It can be shown that $\dot{\phi}_{rm} = 0$ is not an attractor, unless sliding mode occurs, that is, $\sigma = 0$ [25]. From (47), it can be deduced that the term $\mathscr{F}$ and thus the lateral acceleration commands will not encounter any singularity even when $\dot{\phi}_{rm} \to 0$. This is mainly because $\xi < 2$, which implies $2 - \xi > 0$. The bound on gains $\mathscr{M}_i \forall i = 1, 2, 3$ may be chosen by considering the lethal radius of warhead. This implies that the gain requirement will not be unbounded, close to interception. Furthermore, by choosing $\mathscr{M}_i$ and other parameters suitably, we can ensure that ϕ_{rm} converges to a vicinity of zero before $t = r_{am}(t = 0)/(V_a + V_m)$.

Moreover, it is important to note that since the proposed guidance strategy is based on SMC, it has inherent robustness against uncertainties and disturbance rejection properties. As long as sliding mode is enforced and ϕ_{rm} is close to zero, the defender will always remain on the aircraft–attacker LOS, thereby intercepting the attacker before it can intercept the aircraft. Another attractive feature of the proposed strategies is that for both one-way and two-way cooperation and for all values of p, the lateral accelerations demanded by both the defender and the aircraft remain bounded even when the defender is close to intercepting the attacker.

IV. SIMULATION STUDY

In this section, the performance of the proposed guidance strategies is investigated through numerical simulations for various engagement scenarios. The values of gains are chosen as $\mathscr{M}_1 = 0.25$ and $\mathscr{M}_2 = 0.1$. The constant η is chosen to be equal to 10 to ensure small convergence time of error and its rate, once sliding mode occurs in the system. Note that the bounds on the gains derived in Theorems 1 and 2 are based on sufficient conditions, and hence, they are conservative, in general. However, as can be seen from (11), the value of the disturbance term, d, is usually much smaller than its upper bound $d_{\max}$. Hence, even smaller gains such as the ones chosen here are sufficient for interception of the attacker, as will be shown in the following simulations. Furthermore, for a lethal radius of warhead given by $r_{dm} = r_{\text{lethal}}$, the bounds given by (17) and (26) can be further refined. The diamond and circle markers in the trajectories plots denote the initial positions of the aircraft and the attacker, respectively. To avoid problems caused by chattering, we use the sigmoid function instead of the signum function in our simulations, as in [26].

Simulation parameters for all the scenarios are now specified. The attacker and the defender are assumed to have a speed of 200 m/s, while the aircraft has a speed of 100 m/s. The limit on both defender's and attacker's lateral accelerations is assumed to be $20g$, while that of the aircraft is $10g$. The origin is at the aircraft's and defender's initial position. The initial range of attacker with respect to the aircraft is considered to be 5 km with an LOS angle of $\lambda_{am} = 0^\circ$. The initial heading angle of the attacker is $\gamma_m = 150^\circ$, while those of the aircraft and the defender are 45° and 0°, respectively. Since the defender is launched from the aircraft, the initial LOS angle and the relative distance of both the defender and the aircraft from the attacker are the same. However, these might be different for the target and the defender in case the latter is launched from some other platform. The attacker uses PN guidance (PNG) strategy with navigation constant equal to 5. However, this information is unknown to the defender and the aircraft.

A. One-Way Cooperation

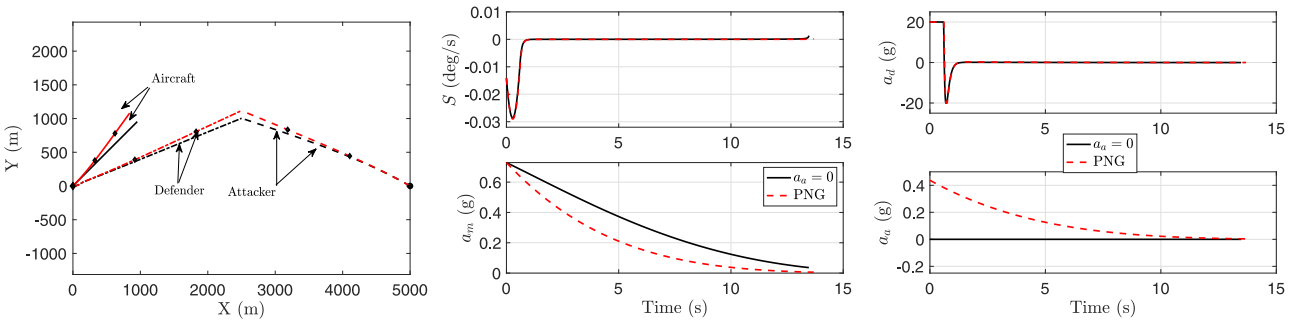
In this subsection, the performance of the guidance strategy, where the aircraft is allowed to make any maneuver without actively cooperating with the defender, is illustrated. However, the aircraft passes the information about its own maneuver to the defender which the latter then uses for guidance computations. Simulations are performed for the two cases, where the aircraft is either nonmaneuvering or uses the PNG strategy. The results are shown in Fig. 5, which depicts trajectories of adversaries, deviations of switching surface, and lateral accelerations of the adversaries.

From Fig. 5, it is evident that the defender is able to intercept the attacker and accomplish its mission in either case. As can be seen from Fig. 5(c), the defender needs more acceleration at the start of engagement to enforce sliding mode and nullify the errors, and then, the acceleration decreases to zero toward endgame. After occurrence of sliding mode, the defender lies on the attacker–aircraft LOS. This ensures the interception of attacker, as shown in Fig. 5(a), before it reaches the aircraft. A second peak in the lateral acceleration profiles can also be observed in Fig. 5(c), which is required to align the defender toward the attacker. It is evident from (17) and (26) that the requirements on gains increase as r_{dm} decreases, and a constant value of gain may not be sufficient throughout engagement, especially for small r_{dm}, and it may result in small deviation of switching surface from zero. This is also visible in the evolution of switching surface in Figs. 5(b), 6(b), and 7(b).

B. Different Levels of Cooperation

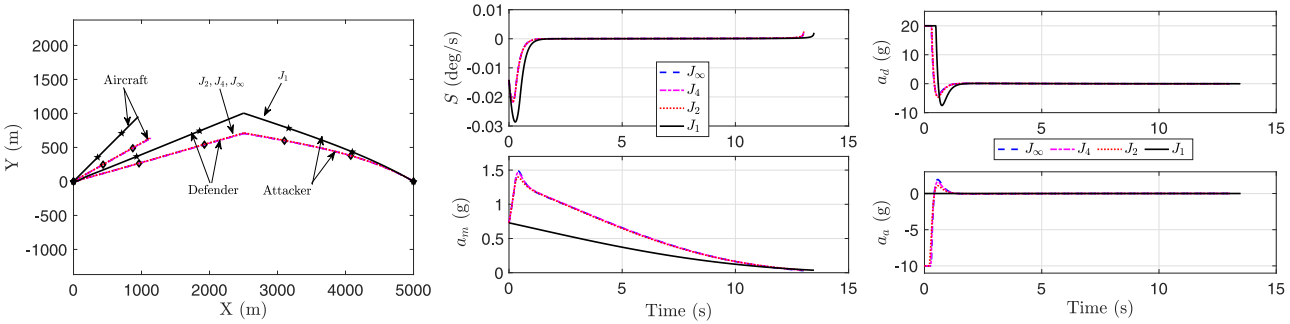
In this subsection, the performances of guidance strategies, where the aircraft actively cooperates with the defender, by minimizing some measure of control effort, are demonstrated. We consider both kinds of switching surfaces: with asymptotic and finite-time convergence. Simulations have been carried out for the four different levels of cooperation, corresponding to the cost function J_p with $p = 1, 2, 4$, and $p \to \infty$.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.



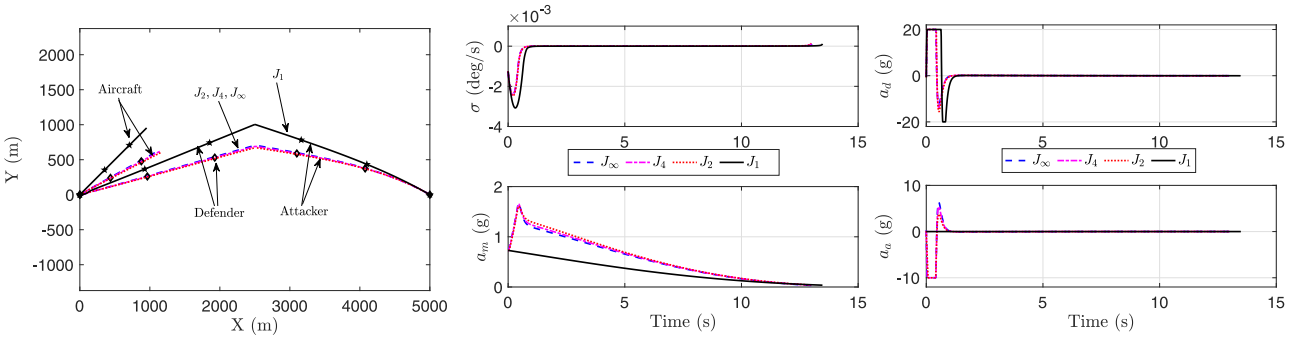
(a) (b) (c)

Fig. 5. Guidance strategy with one way cooperation. (a) Trajectories in (X, Y) space. (b) Deviation of switching surface and attacker lateral acceleration. (c) Defender and aircraft lateral accelerations.



(a) (b) (c)

Fig. 6. Guidance strategy with different levels of cooperation. (a) Trajectories in (X, Y) space. (b) Deviation of switching surface and attacker lateral acceleration. (c) Defender and aircraft lateral accelerations.



(a) (b) (c)

Fig. 7. Finite-time convergent guidance strategy with different levels of cooperation. (a) Trajectories in (X, Y) space. (b) Deviation of switching surface and attacker lateral acceleration. (c) Defender and aircraft lateral accelerations.

1) *Asymptotic Convergence:* The results for switching surface, S, having asymptotic convergence, are shown in Fig. 6, which depicts trajectories of adversaries, deviations of switching surface, and lateral accelerations of the adversaries. From Fig. 6, it follows that the defender–aircraft team intercepts the attacker successfully in all four cases. However, the requirements on their lateral accelerations vary in these cases. For the cost function J_1, the sliding mode occurs later as compared to the cost functions J_2, J_4, and J_∞. This means that the defender takes some more time to reach the aircraft–attacker LOS in case of J_1. Note that for the cost function J_1, the entire burden of interception falls on the defender, and the aircraft does not apply any maneuver, as shown in Fig. 6(c). However, for the other three cases, the defender's requirement is considerably lower. The guidance command profiles for J_2, J_4, and J_∞ are almost identical, as may be observed from Fig. 6. Hence, it appears that for $p > 1$, the nature of cooperation between the defender and the aircraft is similar.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

Comparing Figs. 5(c) and 6(c), we may conclude that the defender pulls out of saturation sooner when there is a two-way cooperation between the aircraft and the defender. This can be attributed to the fact that in two-way cooperation, the aircraft actively aids the defender in reaching the LOS between the attacker and the aircraft. However, note that corresponding to the minimization of the cost function J_1, this benefit cannot be harnessed because the aircraft lateral acceleration remains zero throughout the engagement.

2) *Finite-Time Convergence:* The performances of guidance strategies, with finite-time convergent switching surface, σ, are now investigated. The values of constants m and n are chosen to be 71 and 69, respectively, while the parameters ζ and $\mathscr{M}_3$ equal 10 and 0.25, respectively. The results are shown in Fig. 7, which demonstrates the successful interception of the attacker by the defender while minimizing the different cost functions in (30). Observe that the framework presented in this article can, therefore, accommodate different forms of the variable $\mathscr{F}$ as in (25) and (47). Hence, the proposed strategy ensures successful interception, irrespective of the approach adopted to ensure the convergence of the defender on the LOS between the aircraft and the attacker.

V. CONCLUSION

In this article, we proposed several guidance strategies for the defender and aircraft team considering one-way cooperation, as well as different levels of cooperation that were based on different cost functions. We also considered a nonlinear framework and designed guidance strategies using SMC theory, to ensure successful interception. This obviates the requirement of explicit knowledge about the attacker's maneuvers while only requiring knowledge about the bound on attacker's lateral acceleration capability. Furthermore, the proposed strategies enable guidance command generation without explicit knowledge of time-to-go. Thus, a significant hurdle for many optimal-control-based guidance strategies can be bypassed. Our simulations seem to indicate that for $p > 1$, the optimal lateral accelerations of target–defender team, corresponding to J_p, do not vary significantly, and hence, the nature of aircraft–defender cooperation levels remains similar. Obtaining analytical results along this direction may be a topic for further investigations. The exploration of optimization techniques, which prevent the defender and aircraft lateral accelerations from reaching their saturation limits, is one of the possible future directions of research.

REFERENCES

[1] L. R. Boyell
Defending a moving target against missile or torpedo attack
IEEE Trans. Aerosp. Electron. Syst., vol. AES- 12, no. 4, pp. 522–526, Jul. 1976.

[2] N. A. Shneydor
Comments on defending a moving target against missile or torpedo attack
IEEE Trans. Aerosp. Electron. Syst., vol. 13, no. 3, pp. 321, May 1977.

[3] L. R. Boyell
Counterweapon aiming for defence of a moving target
IEEE Trans. Aerosp. Electron. Syst., vol. AES-16, no. 3, pp. 402–408, May 1980.

[4] T. Yamasaki and S. N. Balakrishnan
Triangle intercept guidance for aerial defense
In *Proc. AIAA Guid. Navigat. Control Conf.*, 2010, Art. no. AIAA 2010–7876.

[5] T. Yamasaki, S. N. Balakrishanan, and H. Takano
Modified command to line-of-sight intercept guidance for aircraft defense
J. Guid., Control, Dyn., vol. 36, no. 3, pp. 901–905, 2013.

[6] T. Yamasaki and S. N. Balakrishnan
Intercept guidance for cooperative aircraft defense against a guided missile
IFAC Proc. Vol., vol. 43, no. 15, pp. 118–123, 2010.

[7] A. Ratnoo and T. Shima
Line-of-sight interceptor guidance for defending an aircraft
J. Guid., Control, Dyn., vol. 34, no. 2, pp. 522–532, 2011.

[8] A. Ratnoo and T. Shima
Guidance strategies against defended aerial targets
J. Guid., Control, Dyn., vol. 35, no. 4, pp. 1059–1068, 2012.

[9] I. Rusnak
The lady, the bandits and the body-guards—A two team dynamic game
IFAC Proc. Vol., vol. 38, no. 1, pp. 441–446, 2005.

[10] I. Rusnak
Guidance laws in defense against missile attack
In *Proc. IEEE 25th Conv. Elect. Electron. Eng. Isr.*, Eilat, Israel, 2008, pp. 90–93.

[11] S. Rubinsky and S. Gutman
Three-player pursuit and evasion conflict
J. Guid., Control, Dyn., vol. 37, no. 1, pp. 98–110, 2013.

[12] I. Rusnak, H. Weiss, and G. Hexner
Guidance laws in target–missile–defender scenario with an aggressive defender
IFAC Proc. Vol., vol. 18. no. 1, pp. 9349–9354,2011.

[13] A. Perelman, T. Shima, and I. Rusnak
Cooperative differential game strategies for active aircraft protection from a homing missile
J. Guid., Control, Dyn., vol. 34, no. 3, pp. 761–773, 2011.

[14] E. Garcia, D. W. Casbeer, and M. Pachter
Cooperative strategies for optimal aircraft defense from an attacking missile
J. Guid., Control, Dyn., vol. 38, no. 8, pp. 1510–1520, 2015.

[15] E. Garcia, D. W. Casbeer, Z. E. Fuchs, and M. Pachter
Cooperative missile guidance for active defense of air vehicles
IEEE Trans. Aerosp. Electron. Syst., vol. 54, no. 2, pp. 706–721, Apr. 2018.

[16] T. Shima
Optimal cooperative pursuit and evasion strategies against a homing missile
J. Guid., Control, Dyn., vol. 34, no. 2, pp. 414–425, 2011.

[17] V. Shaferman and T. Shima
Cooperative multiple-model adaptive guidance for an aircraft defending missile
J. Guid., Control, Dyn., vol. 33, no. 6, pp. 1801–1813, 2010.

[18] O. Prokopov and T. Shima
Linear quadratic optimal cooperative strategies for active aircraft protection
J. Guid., Control, Dyn., vol. 36, no. 3, pp. 753–764, 2013.

[19] M. Weiss, T. Shima, D. Castaneda, and I. Rusnak
Combined and cooperative minimum-effort guidance algorithms in an active aircraft defense scenario
J. Guid., Control, Dyn., vol. 40, no. 5, pp. 1241–1254, 2017.

[20] S. R. Kumar and T. Shima
Cooperative nonlinear guidance strategies for aircraft defense
J. Guid., Control, Dyn., vol. 40, no. 1, pp. 124–138, 2017.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.

[21] V. Utkin, J. Guldner, and J. Shi
Sliding Mode Control in Electro-Mechanical Systems (Automation Control Engineering Series), 2nd ed. Boca Raton, FL, USA: CRC Press, 2009, ch. 2, 3, 8, pp. 17–63, 159–203.

[22] A. Sinha and S. R. Kumar
Super-twisting control based cooperative salvo guidance using leader-follower approach
IEEE Trans. Aerosp. Electron. Syst., vol. 56, no. 5, pp. 3556–3565, Oct. 2020.

[23] S. R. Kumar and D. Mukherjee
Cooperative salvo guidance using finite-time consensus over directed cycles
IEEE Trans. Aerosp. Electron. Syst., vol. 56, no. 2, pp. 1504–1514, Apr. 2020.

[24] S. R. Kumar and D. Mukherjee
Cooperative guidance strategies for active aircraft protection
In *Proc. Amer. Control Conf.*, 2019, pp. 4641–4646.

[25] Y. Feng, X. Yu, and Z.Man
Non-singular terminal sliding mode control of rigid manipulators
Automatica, vol. 38, no. 12, pp. 2159–2167, 2002.

[26] S. R. Kumar, S. Rao, and D. Ghose
Sliding-mode guidance and control for all-aspect interceptors with terminal angle constraints
J. Guid., Control, Dyn., vol. 35, no. 4, pp. 1230–1246, 2012.



Shashi Ranjan Kumar received the B.Sc. (Engg.) degree in electronics and communication engineering from the Muzaffarpur Institute of Technology, Muzaffarpur, India, in 2008, and the M.E. and Ph.D. degrees in aerospace engineering from the Indian Institute of Science, Bangalore, India, in 2010 and 2015, respectively.

From 2015 to 2017, he was a Postdoctoral Fellow with the Faculty of Aerospace Engineering, Technion—Israel Institute of Technology, Haifa, Israel. He is currently an Assistant Professor of Aerospace Engineering with the Indian Institute of Technology Bombay, Mumbai, India. His research interests include guidance of aerospace vehicles, path planning, multiagent systems, and cooperative control.



Dwaipayan Mukherjee received the B.E. degree in electrical engineering from Jadavpur University, Kolkata, India, in 2007, the M.Tech. degree in control systems engineering from the Indian Institute of Technology Kharagpur, Kharagpur, India, in 2009, and the Ph.D. degree in aerospace engineering from the Indian Institute of Science, Bangalore, India, in 2014.

From 2015 to 2017, he was a Postdoctoral Fellow with the Faculty of Aerospace Engineering, Technion—Israel Institute of Technology, Haifa, Israel. He is currently an Assistant Professor of Electrical Engineering with the Indian Institute of Technology Bombay, Mumbai, India. His research interests include multiagent systems, cooperative control, and control theory.

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY MADRAS. Downloaded on October 13,2023 at 08:19:00 UTC from IEEE Xplore. Restrictions apply.